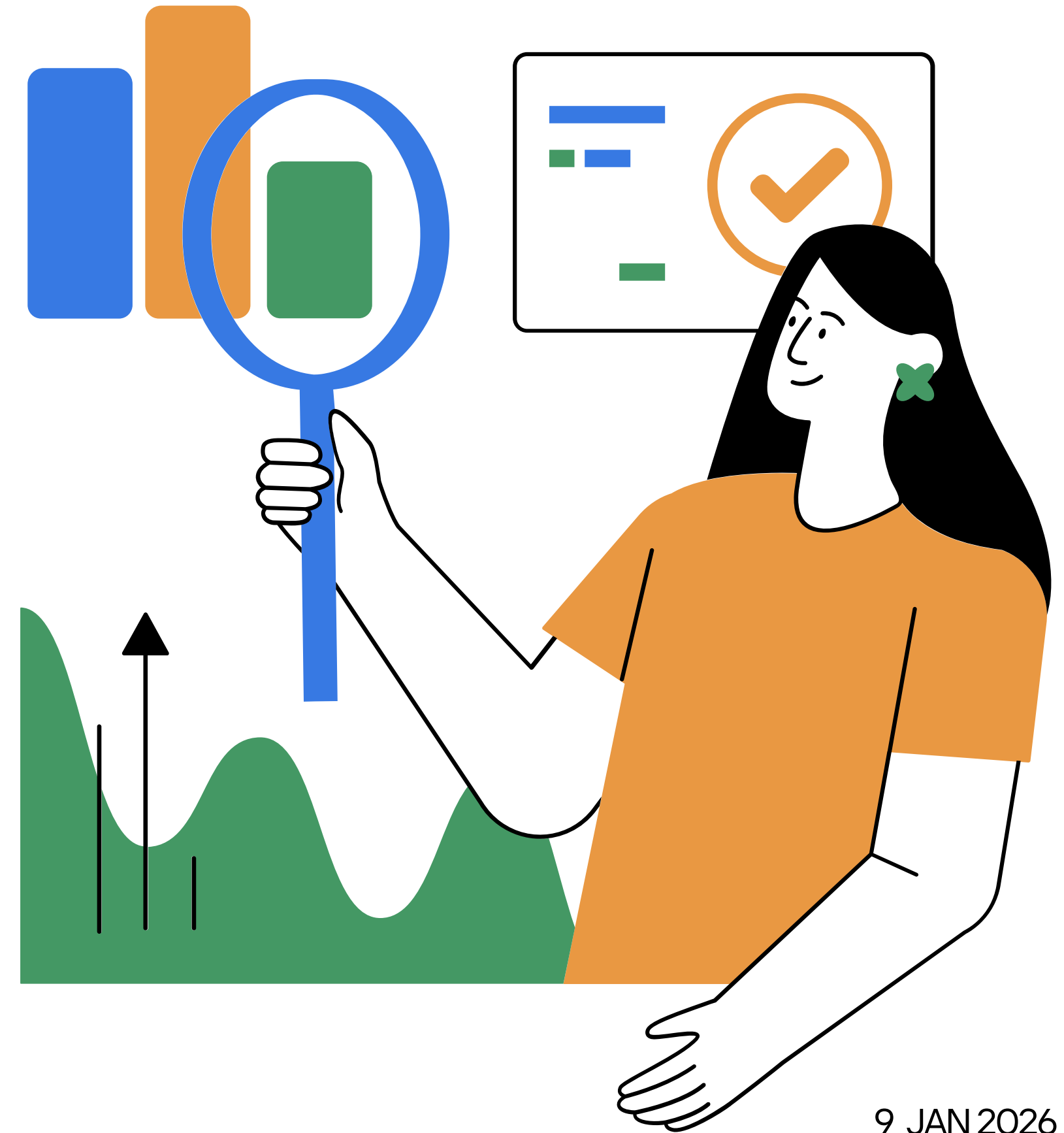
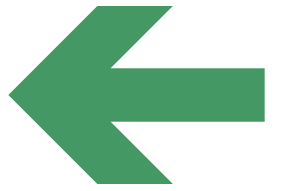


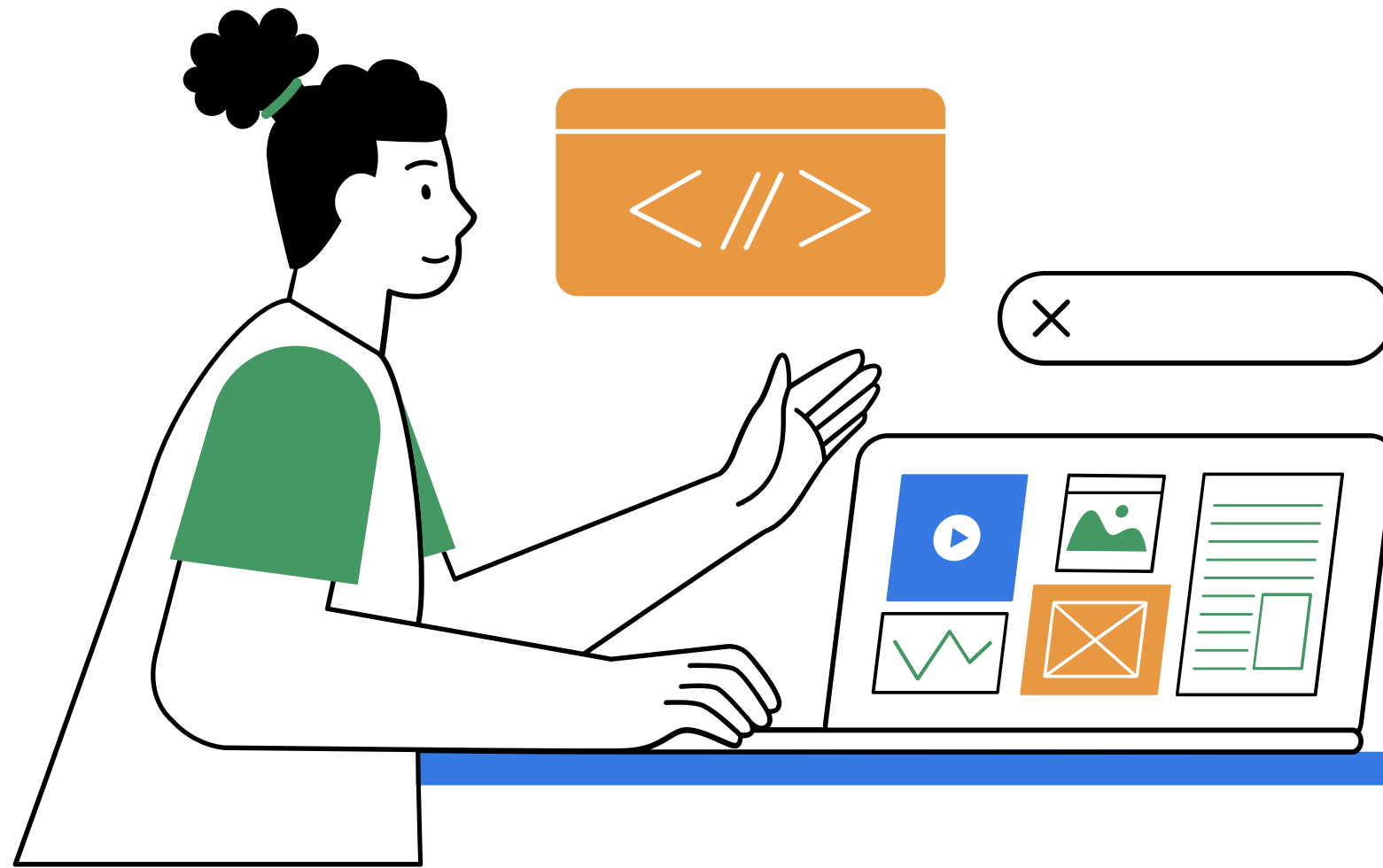
Bank Loan Analysis





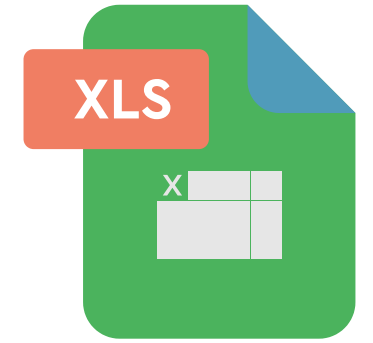
Introduction

- This project focuses on analyzing bank loan data to understand lending trends, customer behavior, and risk distribution.
- Dashboards are created using Power BI, Tableau, and Excel.
- The analysis helps banks improve credit decisions and portfolio performance.



Tools & Technologies

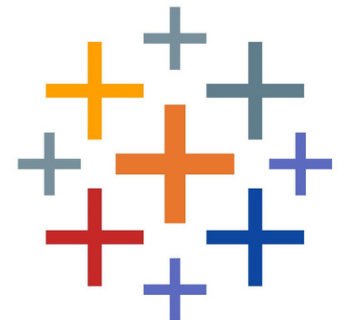
Excel



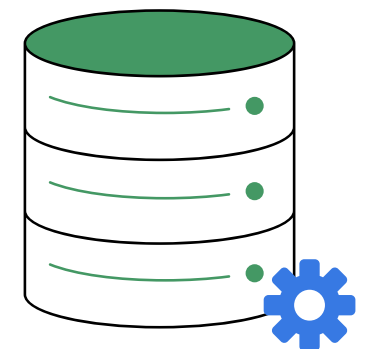
PowerBI



Tableau

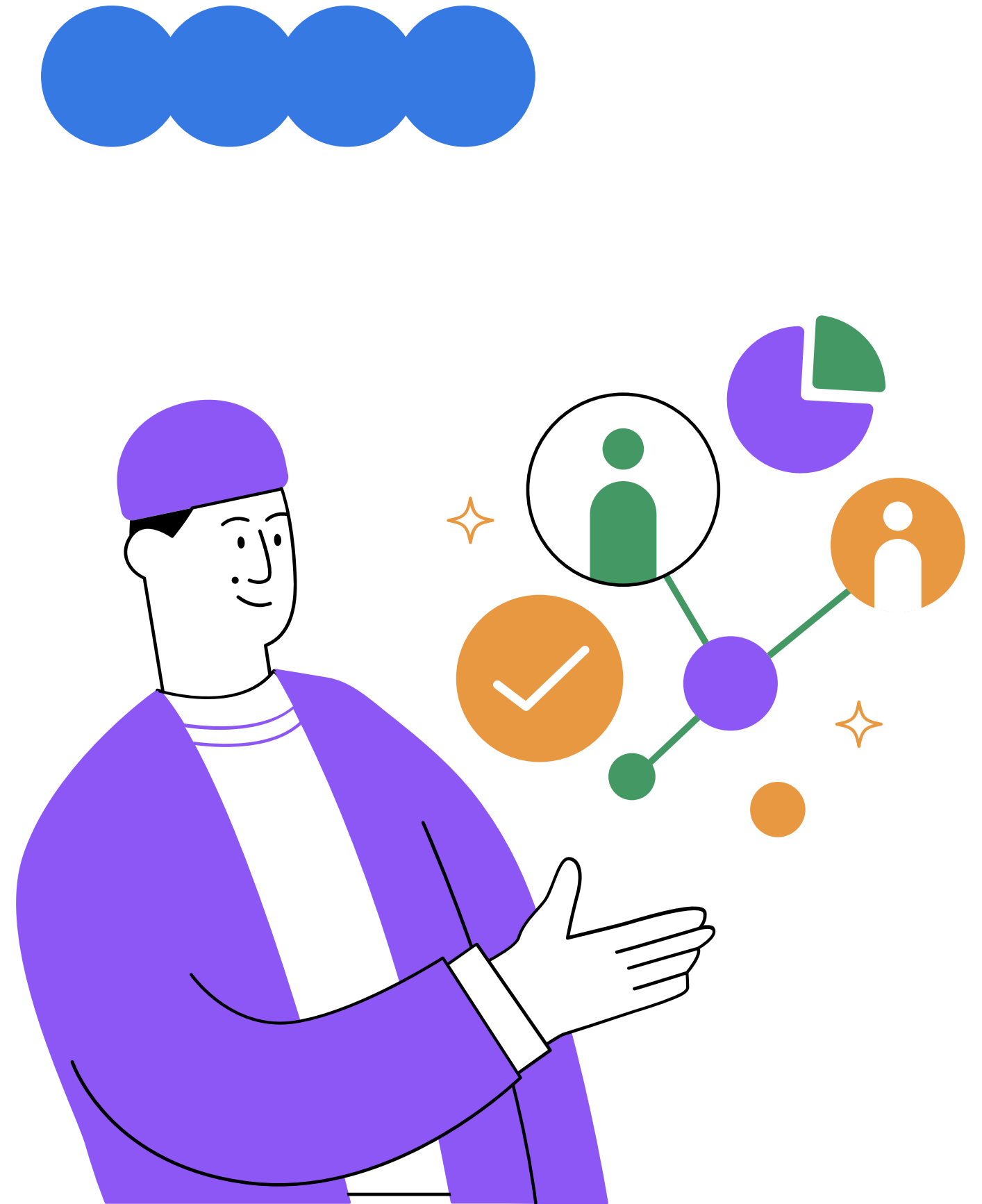


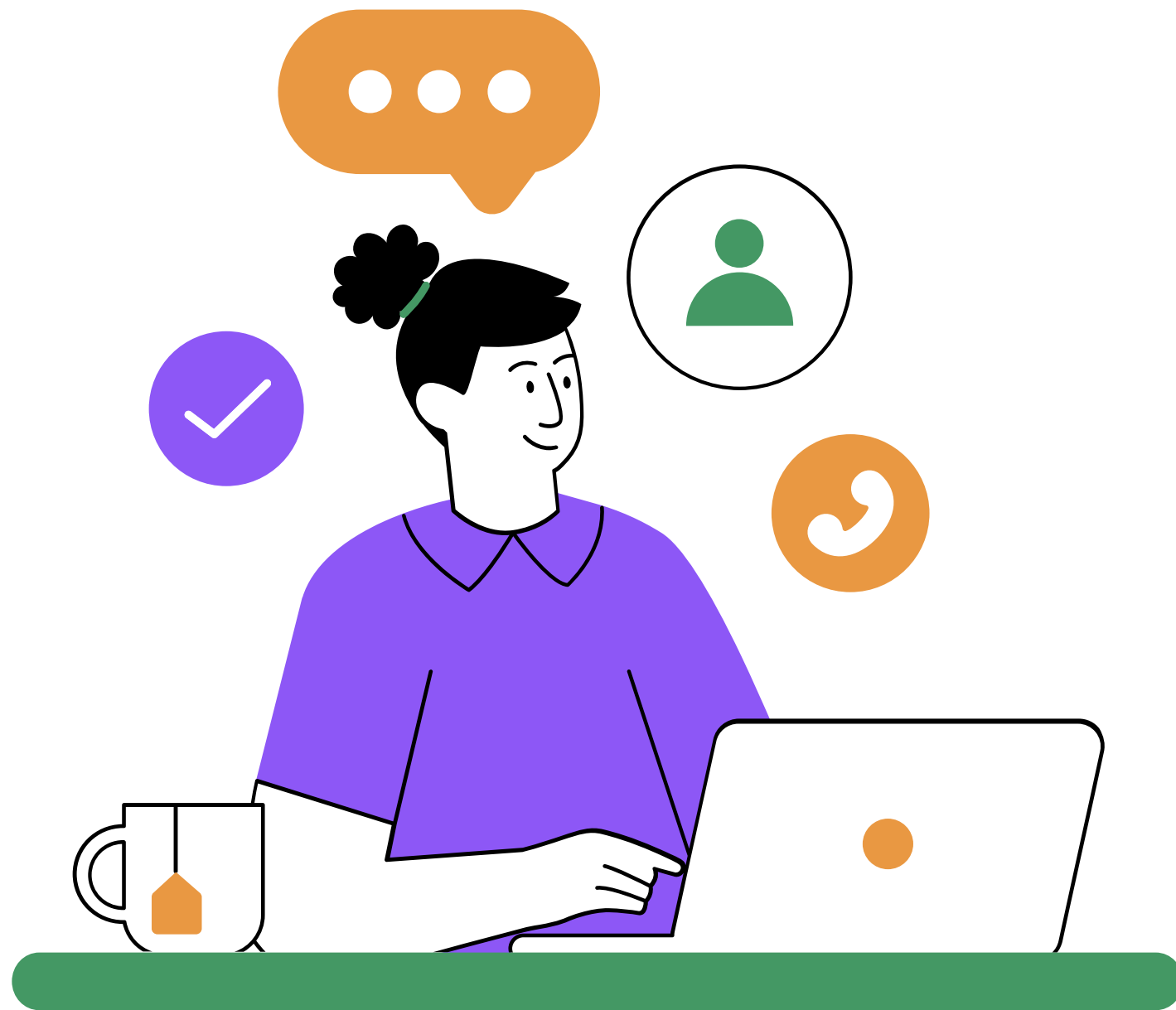
SQL



Objectives

- Analyze year-wise loan disbursement trends
- Understand customer verification and risk grades
- Evaluate state-wise loan distribution
- Study home ownership vs repayment behavior
- Support data-driven banking decisions



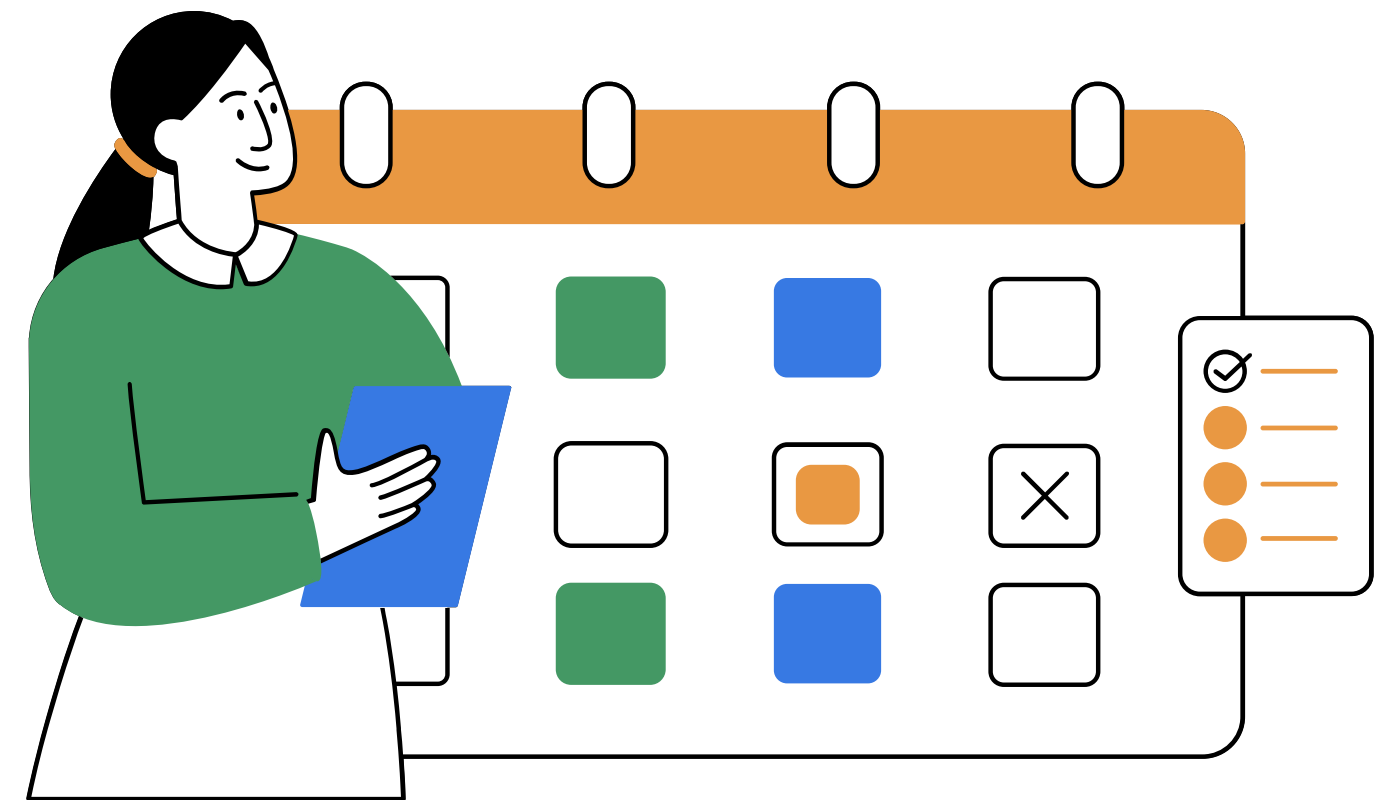


Key Performance Indicators [KPI]

- 💰 Total Loan Amount: ~446 Million
- 👤 Total Customers: ~39.7K
- 📈 Average Interest Rate: ~12.02%
- ⚖️ Average DTI (Debt-to-Income): ~13.32%

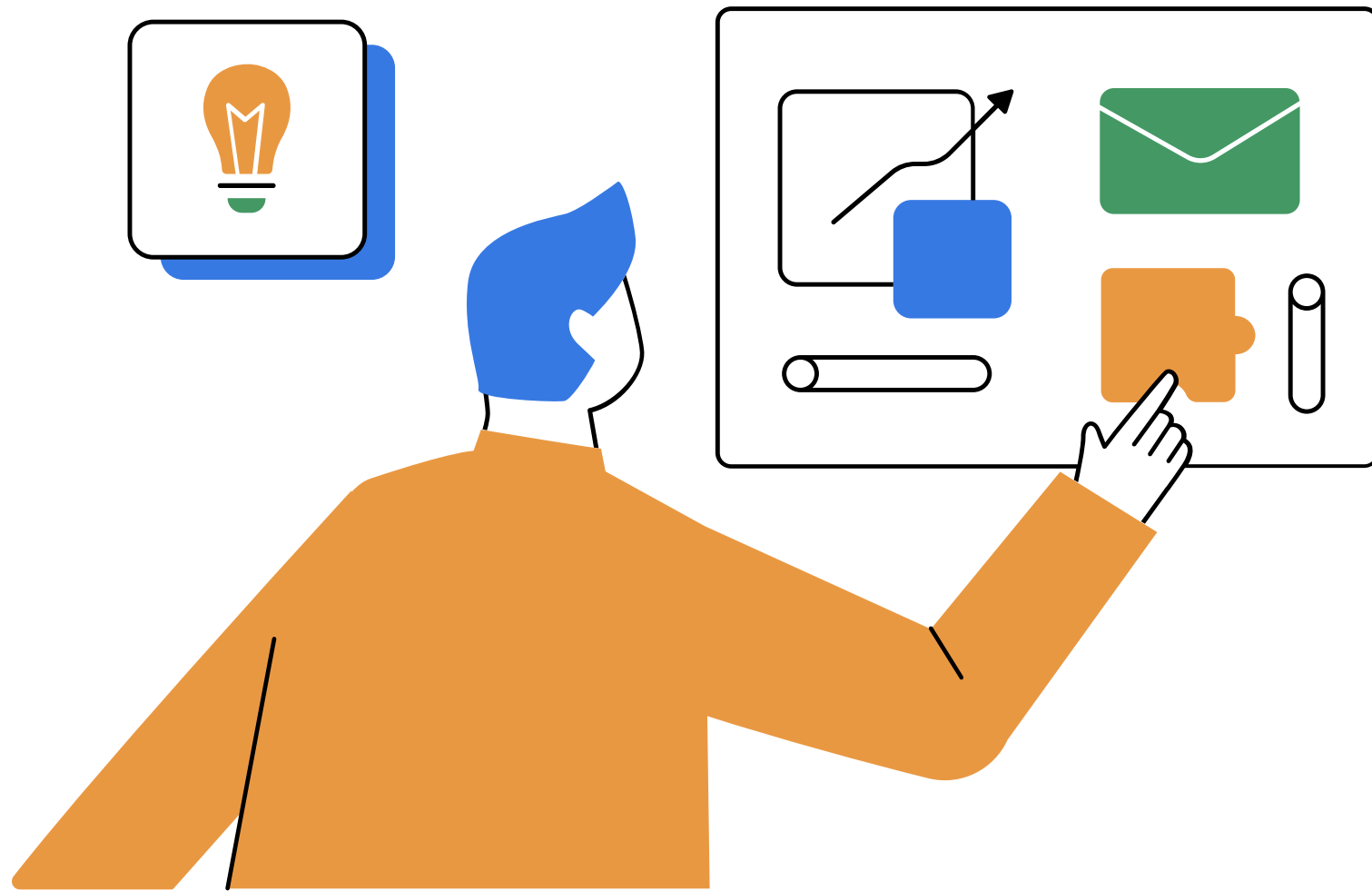
Year-wise Loan Amount Analysis

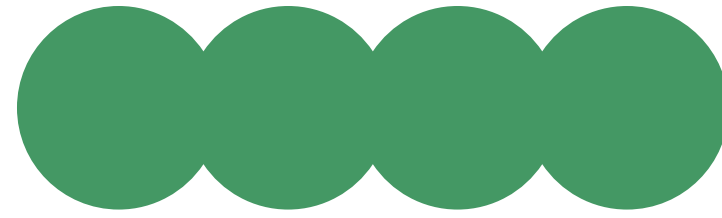
- Loan amount shows strong growth from 2007 to 2011.
- Major increase observed after 2009.
- Indicates rising credit demand and bank expansion.



Grade & Sub-grade Wise Loan Balance

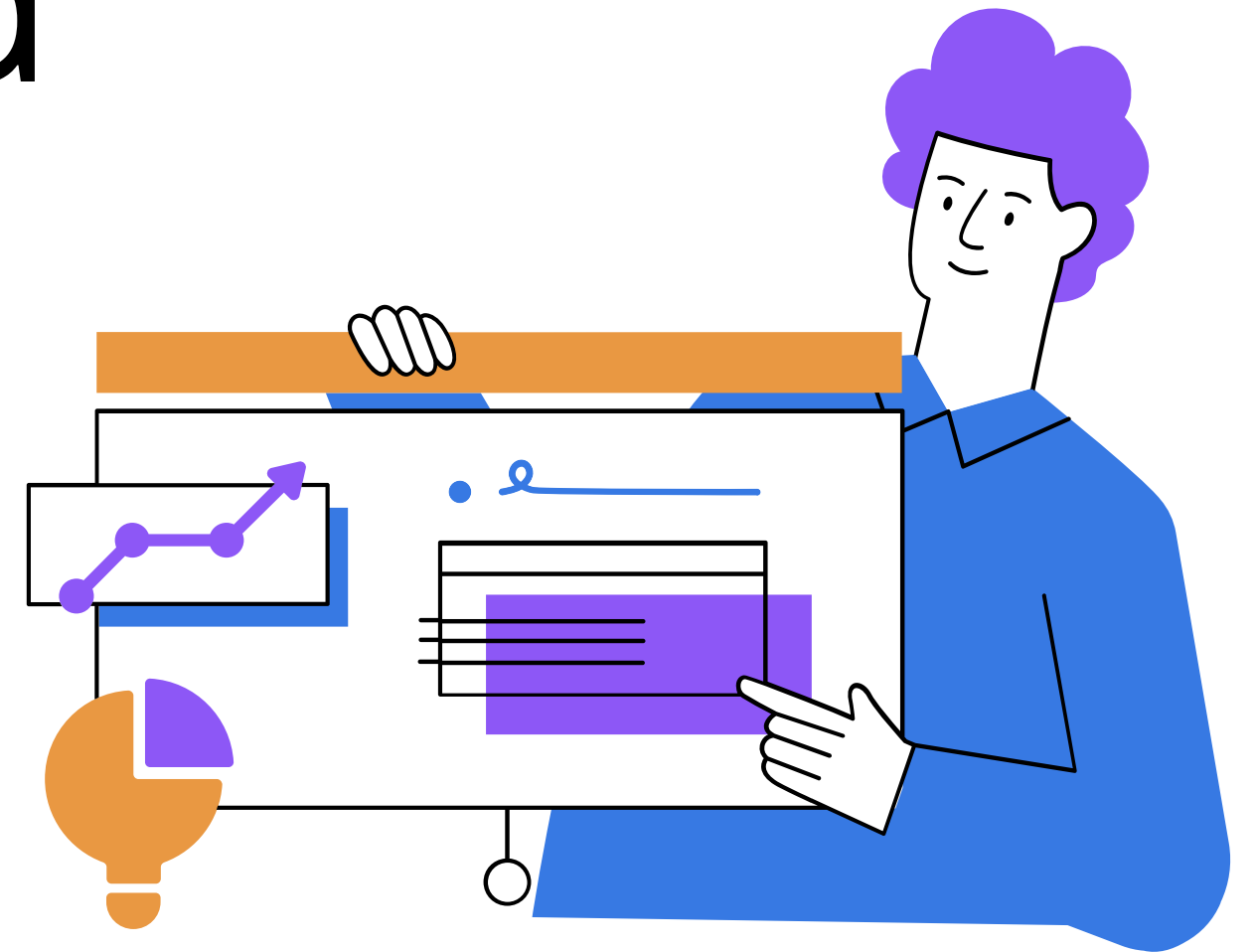
- Grade A and B customers contribute highest loan balances.
- Lower grades show reduced loan exposure.
- Helps in assessing portfolio risk.





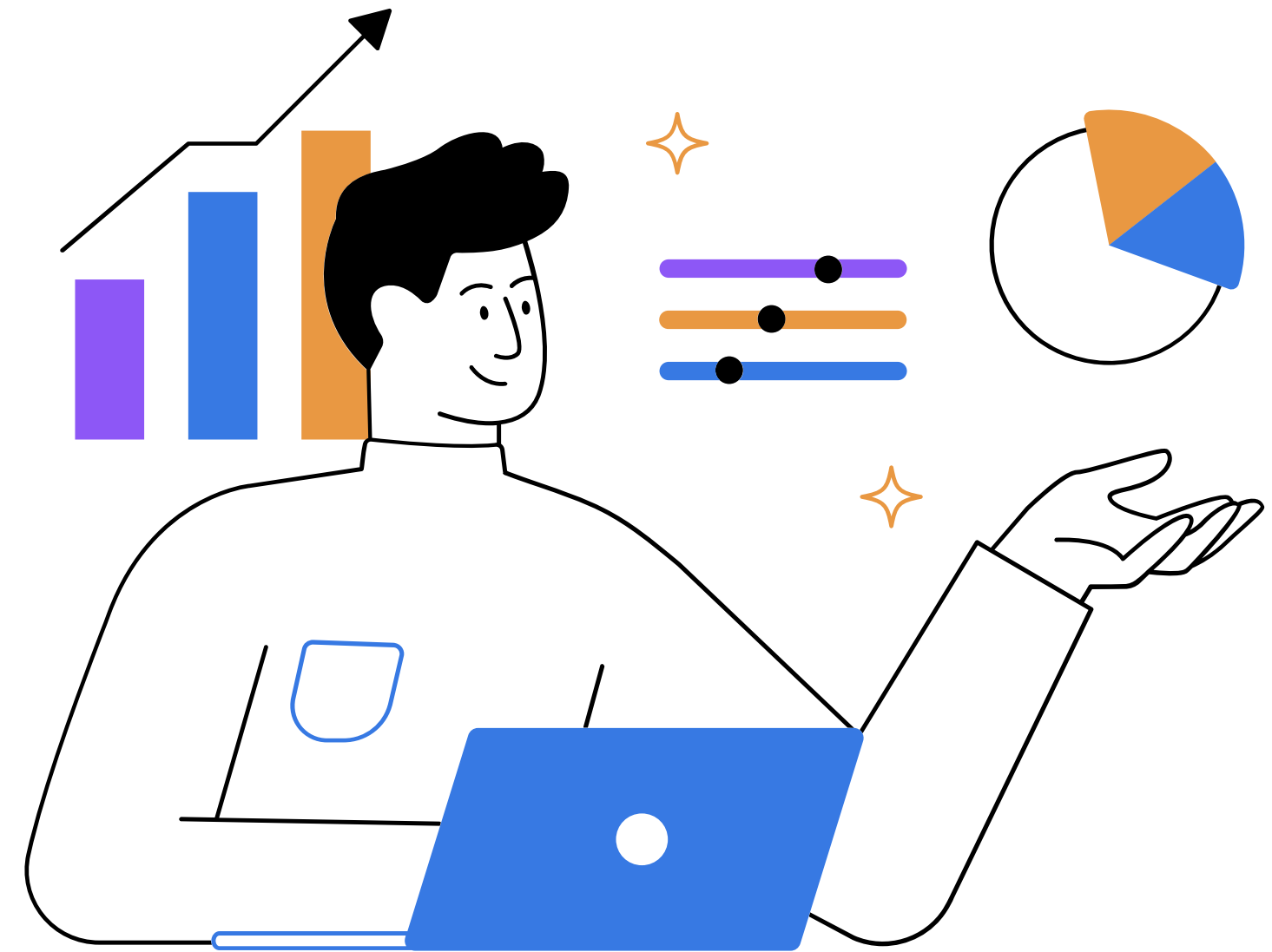
Verified vs Non-Verified Customers

- Majority of loan amount comes from verified customers.
- Verified loans are relatively lower risk.
- Non-verified loans require closer monitoring.



State-wise Loan Distribution

- California has the highest loan amount.
- Other major states show moderate contribution.
- Useful for regional performance analysis.



Home Ownership vs Last Payment Year

- Mortgage customers dominate the loan portfolio.
- Rent and Own categories show varying repayment patterns.
- Supports customer segmentation strategy.

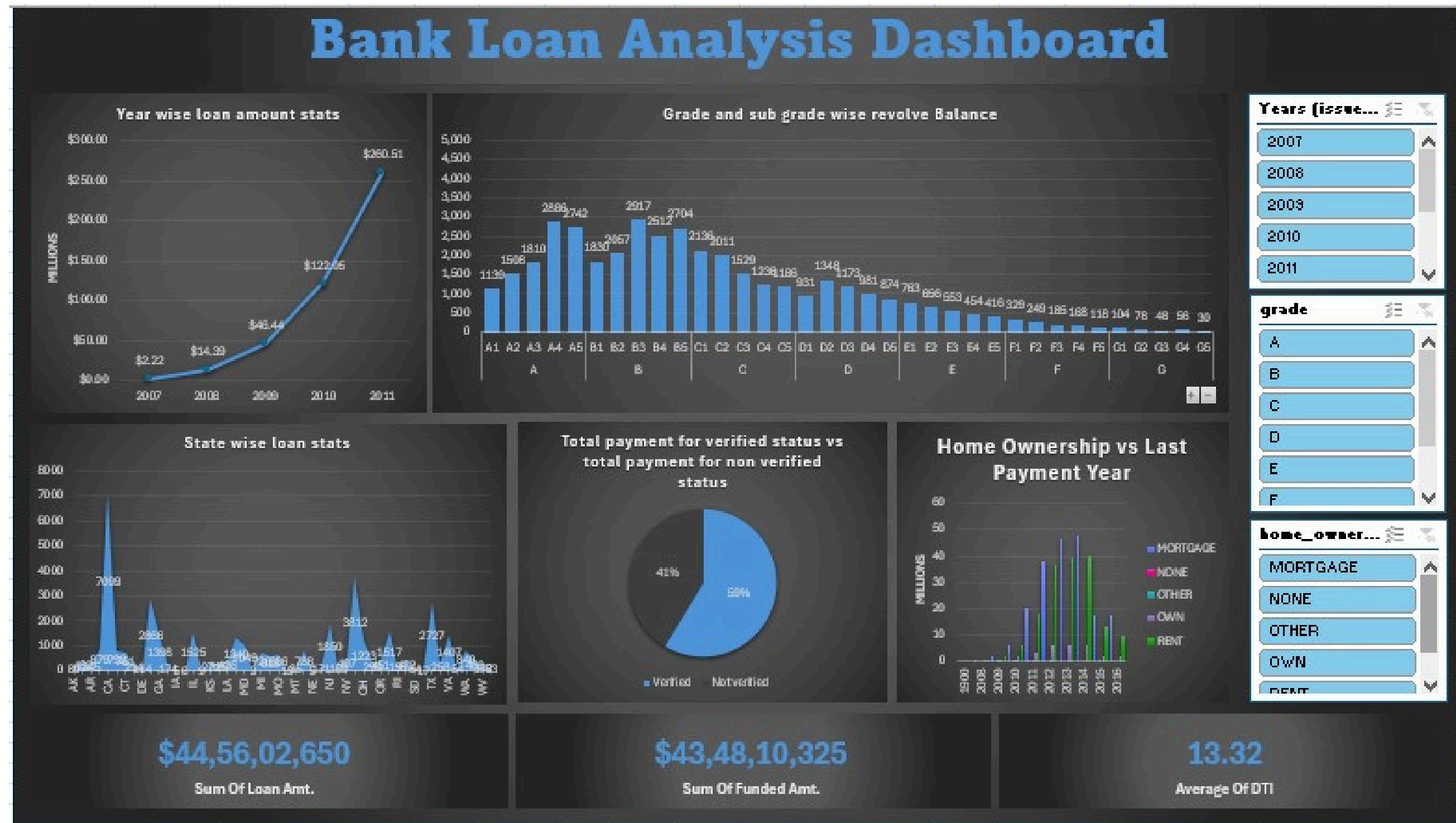


Conclusion

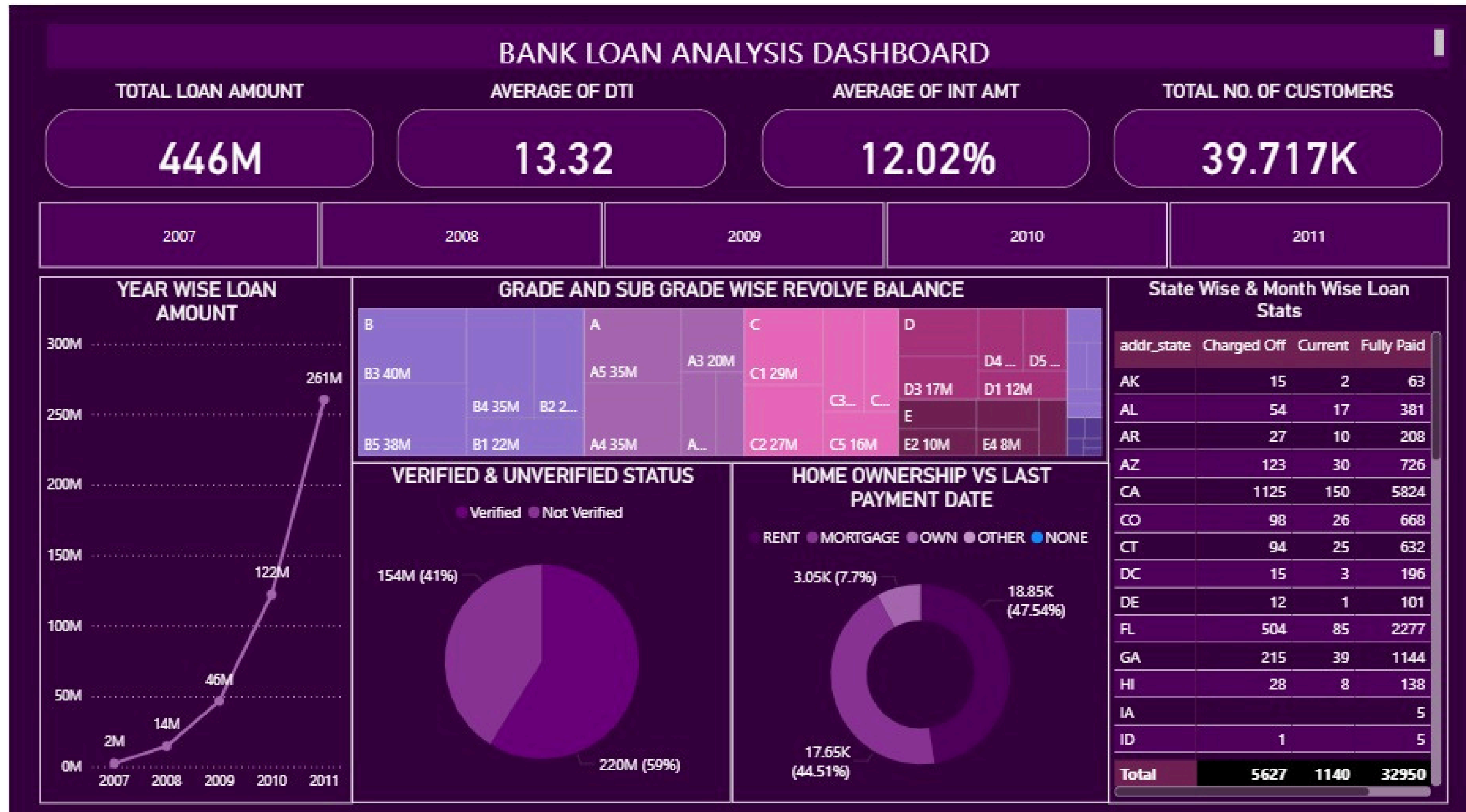
- The Bank Loan Analysis provides valuable insights into lending behavior and risk.
- Helps banks improve credit policies and customer targeting.
- Dashboards enable quick and effective decision-making.



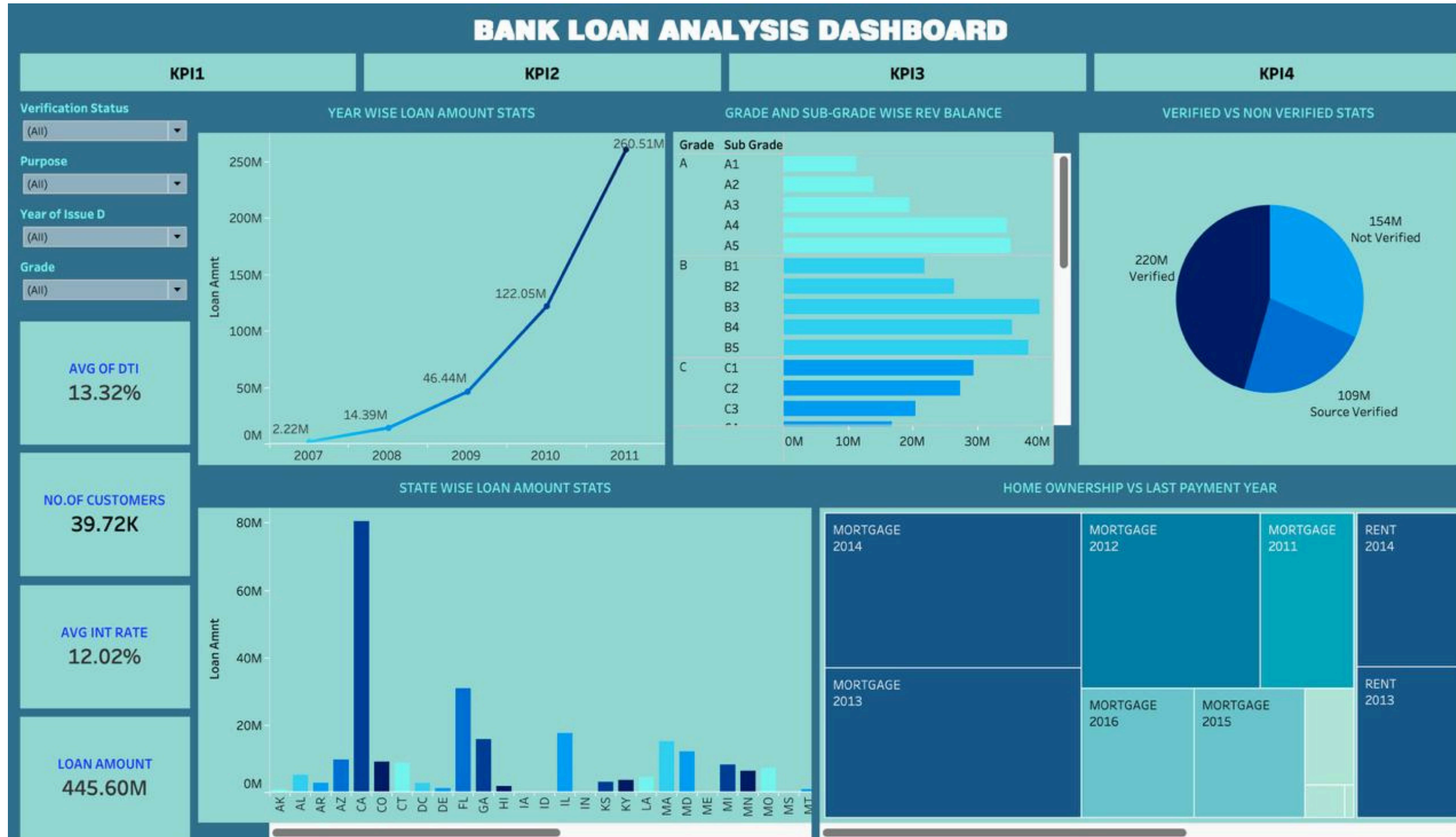
Excel-Dashboard



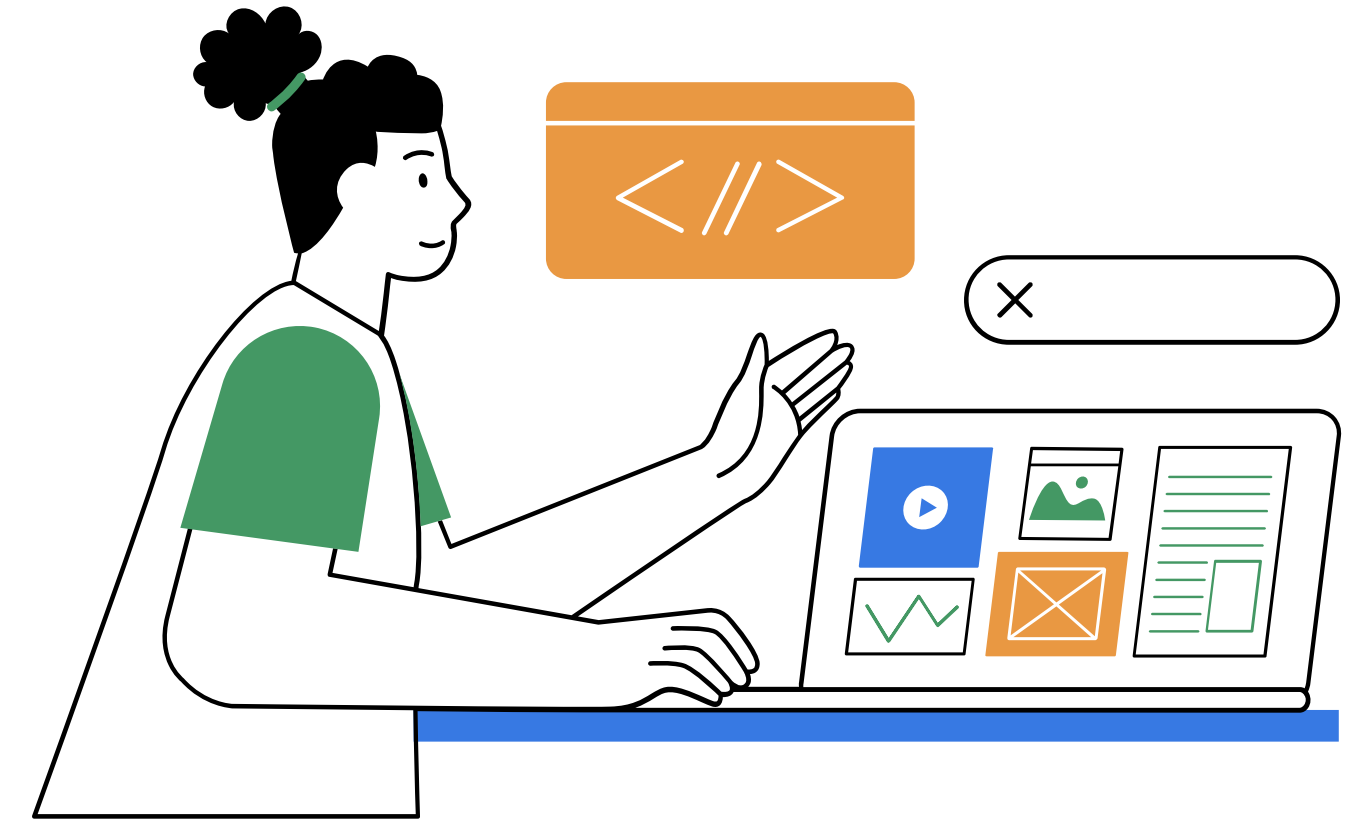
Power BI



Tableau



Bank Loan Analysis



Challenges Faced

- Handling large loan datasets across multiple years
- Data quality issues such as missing and inconsistent values
- Managing multiple dimensions like state, grade, and verification
- Ensuring dashboards remain simple and user-friendly
- Aligning insights with real banking use cases



Key Insights – Overall Performance

- Total Loan Amount exceeds 445 Million
- Customer base of nearly 40,000 borrowers
- Average Interest Rate around 12%
- Average DTI indicates moderate credit risk
- Strong loan growth observed after 2009



Business Value & Use Case

- Helps banks identify high-performing regions and customer segments
- Supports risk assessment through grade-wise analysis
- Improves monitoring of non-verified customers
- Enables data-driven loan approval strategies
- Provides management-ready insights via dashboards



Bank Loan Analysis Dashboard

Risk • Growth • Customer Insights

Presented by: Group 6

Thank You